



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VR - CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

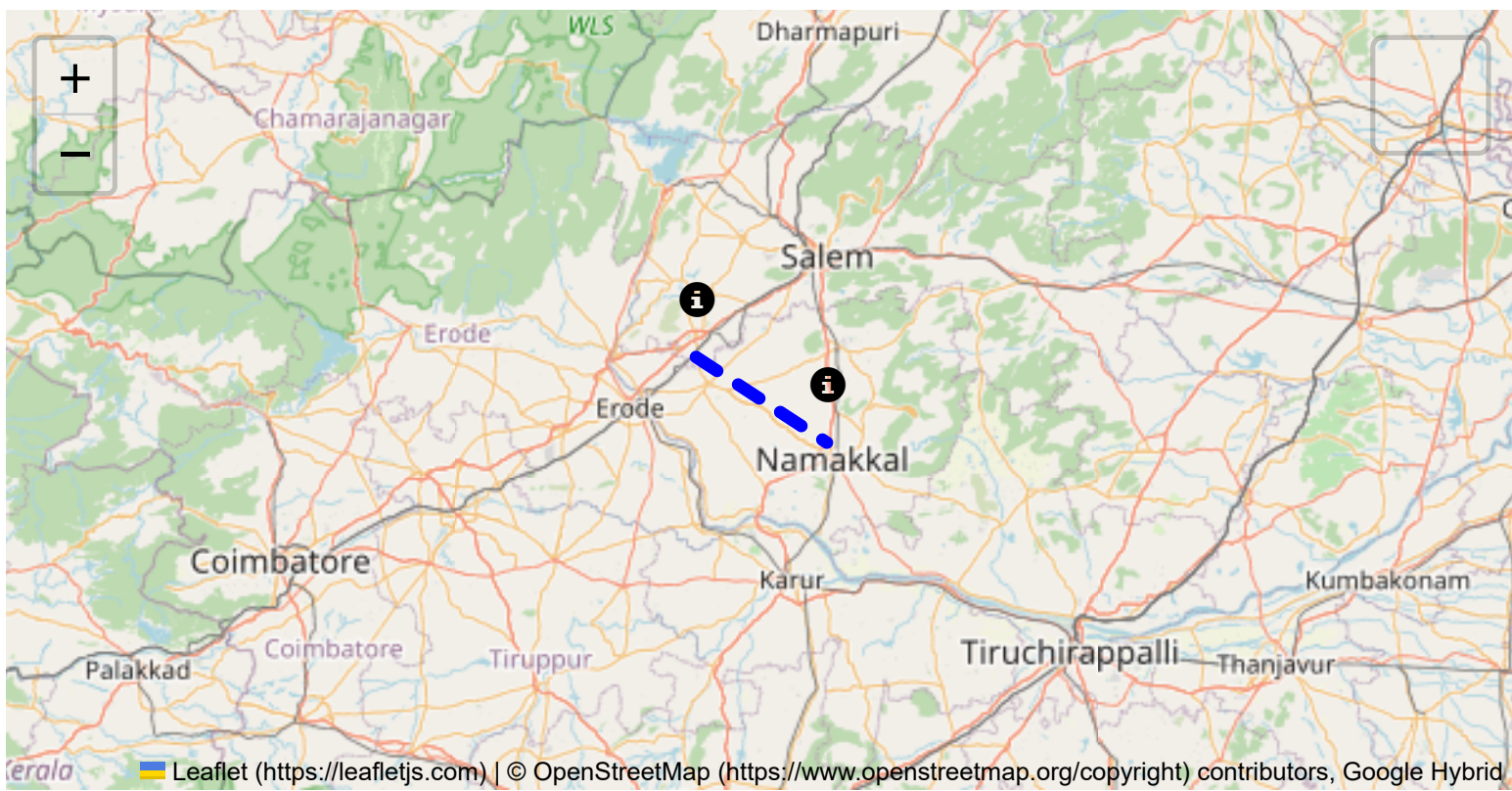
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

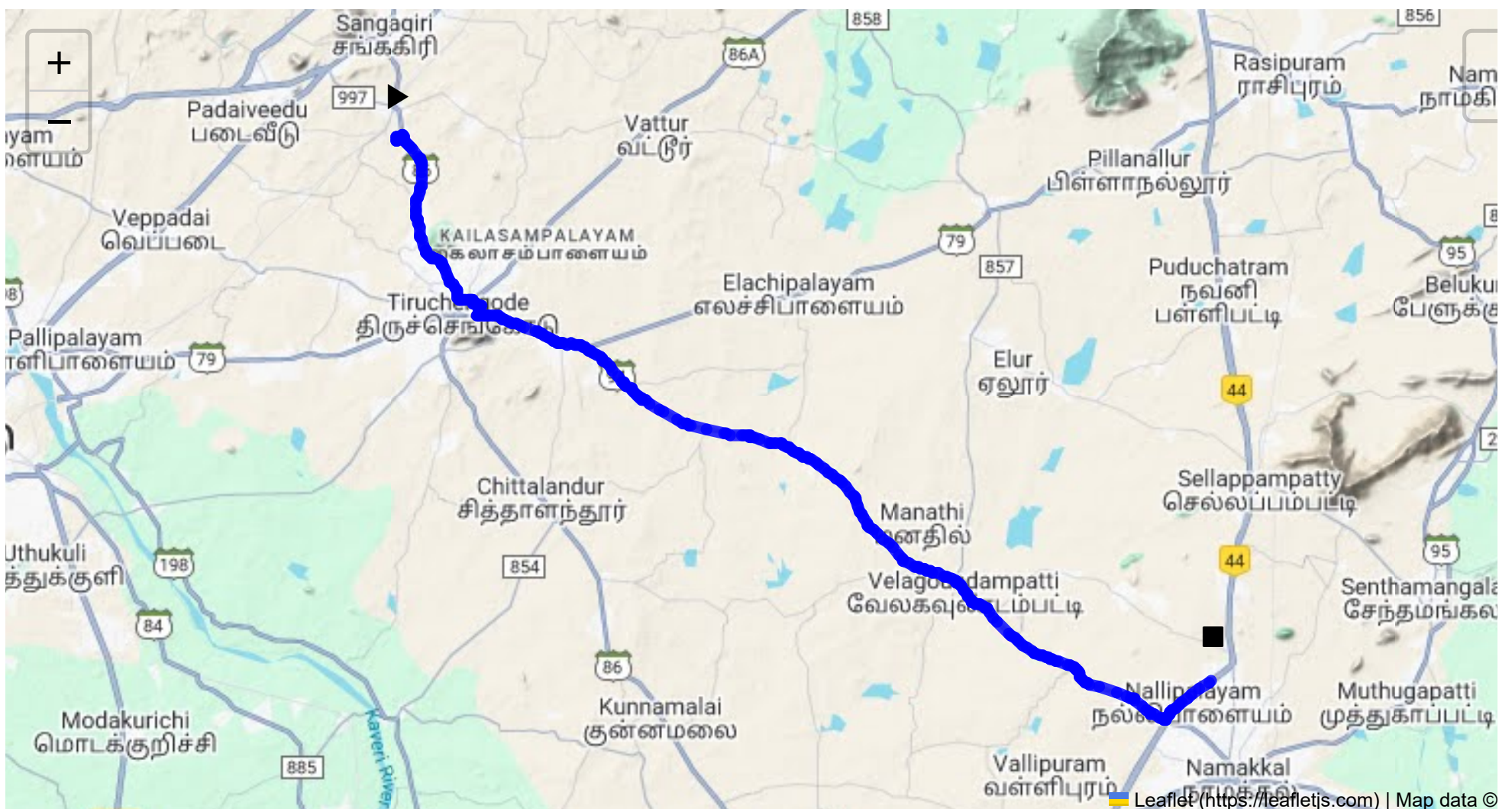
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.85 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.253346, 78.158318)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to NH 44, Service Rd, Swamy Nagar in Namakkal spans approximately 42.85 kilometers. It typically takes around 56 minutes for heavy vehicles carrying hazardous material, following major state highways and local roads that connect these two points.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** Tamil Nadu experiences a tropical climate with distinct dry and wet seasons. The monsoon season from June to September can bring heavy rainfall, which may cause waterlogging and reduce visibility.
- **Potential Hazards:** During the monsoon season, roads can become slippery, and visibility may be compromised, increasing the risk of accidents. Flooding can sometimes lead to road closures.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is more common during morning (8:00 to 10:00 AM) and evening (5:00 to 7:00 PM) rush hours, particularly in urban areas around Namakkal.
- **Congestion-Prone Areas:** The vicinity around Namakkal, particularly near NH 44 and local markets, is known for higher congestion levels.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The roads connecting Sangagiri to Namakkal are generally well-maintained. However, some rural roads may experience potholes and uneven surfaces, especially after heavy rains.
- **Infrastructure:** Adequate signage and road markings are common, but sudden detours and road work may occur without notice.

5. Suggestions for Alternative Routes

For emergencies, detouring via nearby state highways (SH) may provide relief, although these might add distance and are subject to weather conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials requires permits and adherence to specific safety protocols. Restrictions may apply during peak hours in certain transit zones, particularly in densely populated areas.

7. Overview of Historical Incidents

While specific data on historical incidents involving heavy vehicles in this exact route may not be readily available, Tamil Nadu generally observes risks of overturns and collisions in areas with sharp turns or congested traffic, particularly during inclement weather.

8. Environmental Considerations and Sensitive Areas

The route traverses both urban and rural environments. Care must be taken to avoid spills, especially near water bodies and agricultural zones to prevent contamination.

9. Analysis of Communication Coverage

Most areas of the route maintain good cellular network coverage due to proximity to urban centers. However, rural segments might experience occasional dead zones.

10. Estimated Emergency Response Times

- Urban Sections:** Emergency services can typically respond within 20-30 minutes.
- Rural Sections:** Response times can extend to 40-60 minutes due to road conditions and distance from major towns.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Namakkal is moderately safe for heavy vehicles transporting hazardous materials, with potential risks primarily during the monsoon season due to adverse weather conditions. Traffic congestion around Namakkal, road maintenance after rain, and the need for adherence to regulations are critical considerations. Having alternative route plans and proper communication tools to address dead zones will improve travel safety and efficiency. Emergency response times may vary, necessitating that drivers be well-trained in handling incidents involving hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.38346, 77.89975	15 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	High	11.23961, 78.14223	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23961, 78.14223

